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Research article

Climate worry and daily life impacts among a sample of Canadian youth accessing integrated youth services

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A R T I C L E I N F O

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A B S T R A C T

Introduction: Despite calls for service providers to support youth in navigating the mental health impacts of climate change, data from clinical contexts remain limited. Our research explores climate worry among youth accessing integrated youth services (IYS) in British Columbia, Canada.
Methods: Youth aged 16-24 years (n=393) were recruited from the waiting rooms of four IYS centres. Participants completed a survey to capture sociodemographic characteristics and assess mental health, including single items on level of climate change worry and whether feelings about climate change negatively impacted their daily lives.
Results: Overall, 40.9% of participants were very or extremely worried about climate change and 28.0% of participants reported that their feelings about climate change resulted in negative impacts on their daily life. Gender diverse youth and older youth were more likely to report negative daily life impacts due to their feelings about climate change. Moreover, while climate worry was inconsistently associated with measures of mental health, reporting that climate change was negatively impacting daily life was associated with higher scores of depressive symptoms, psychological distress, and worsened functioning across multiple domains.
Conclusion: This study is among the first to examine the relationship between climate change and youth mental health in a clinical sample. From these results, it is clear that a substantial subset of youth accessing clinical services are identifying that their feelings about climate change are negatively impacting their daily lives. Based on these results, service providers should be attuned to the impacts of climate change on youth mental health.

1. Introduction

Climate change and its effects have been linked to a growing number of adverse consequences for human health [1–3]. One health consequence that is receiving increasing attention is the negative impacts of climate change on mental health, particularly among youth [4–6]. These mental health impacts can be either direct or indirect [6,7]. The former refers to impacts resulting from experiences of climate change-induced disasters, such as wildfires, floods, and storms, as well as witnessing more gradual changes, such disruptions to local ecosystems. The latter refers to impacts resulting from the more existential concerns arising from overall awareness of climate change and its impacts [6,7]. As a

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whole, these impacts have been found to exacerbate or trigger mental health concerns, including anxiety, depression, and post-traumatic stress disorder [6,8,9].

Alongside these mental health concerns, climate change has been found to contribute to various challenging emotional responses, including but not limited to fear, anxiety, anger, despair, and grief [10–14]. Various terms have been suggested to describe these responses, including climate anxiety, eco-anxiety, and climate worry, as well as more specialized terms such as solastalgia, ecological grief, Anthropocene horror, and eco-despair [6,10,15–17]. Given this profusion of terminology, *climate distress* has been used as an umbrella term to refer to these impacts collectively [18,19]. Of note, challenging emotions related to climate change are not inherently pathological [11,13,20–22]. Indeed, such reactions can be thought of as rational responses to the state of the world. Some scholars have suggested that they can even be adaptive [10], with positive relationships to pro-environmental behaviours [14,23].

However, research also suggests that, for some youth, experiences of climate distress are associated with mental ill-health [24], with youth considered at increased risk of these negative mental health consequences [25], owing to their stage of brain development [25], higher exposure to distressing media [26], and differing temporal orientation [27]. Furthermore, research has identified that certain youth are more likely to experience climate distress, with girls and gender diverse youth often identified as having higher levels of climate distress [12,28].

Recognizing these risks, there is a growing body of literature exploring the mental health impacts of various forms of climate distress, including climate worry, on youth, mostly at a population-level [18,24]. The overall pattern is that various forms of climate distress are associated with worsened youth mental health [18,24], including climate worry [29–31].

However, data on the mental health impacts of climate change among youth accessing clinical services remain limited [32]. This is concerning given that the studies that are available indicate this issue is leading to clinical presentations [33,34]. While it has been suggested that climate-change related emotional responses may reach the level of clinical significance when an individual is unable to able to cope with them on their own [24] or when there are impacts to daily functioning [35], there remain gaps in understandings about when clinical intervention might be warranted [32].

Given these knowledge gaps, the purpose of this paper is to understand how climate worry and daily life impacts due to emotions about climate change present in a sample of youth aged 16-24 years accessing health and social services in British Columbia (BC), Canada. The primary aim is to determine to what extent climate worry and negative daily life impacts due to emotions about climate change in a clinical sample are associated with measures of mental health and functioning. A secondary aim is to identify youth characteristics that may increase the likelihood of experiencing emotions about climate change that negatively impact their daily life.

2. Methods

This study presents findings on the mental health impacts of climate change for youth who accessed clinical services at Foundry, an integrated youth services (IYS) initiative in BC, Canada, between March 2023 and April 2024. Foundry consists of a network of community-based centres, and a virtual service, that co-locates low-barrier mental health, substance use, physical/sexual health, peer support, and social services [36,37]. BC has experienced several climate events in recent years, such as a Heat Dome in 2021, which killed 619 people [38]. Furthermore, the interior region of BC experiences frequent, destructive wildfires [39], which result in air quality advisories across many areas of the province [40,41].

As part of a larger study examining youth health and functioning, youth aged 16-24 years were recruited from the waiting rooms of four Foundry centres; three centres are located in the Lower Mainland region of BC, and one is located in the Northern region of the province. Youth who were waiting to receive clinical services were approached by a member of the study team and informed about the opportunity to participate in the study.

Ethical approval was attained from the University of British Columbia Behavioural Research Ethics Board (H20-03100). Informed consent was obtained from all participants. The consent form included information about the study, potential risks, how the data would be used, and who the youth could contact with further questions or concerns. Youth were informed that their data would be kept confidential and not shared with their healthcare team. Youth were provided with a tablet computer to complete the survey. Participants were compensated for their time with a $10 cash honorarium and entered into a gift card draw. They were also provided with information on locally available mental health resources.

2.1. Measures

Sociodemographic questions characterized the sample according to a range of personal characteristics, including age, gender, housing status, and education.

Mental Health was assessed using a variety of measures, including a single-item measure of overall mental health: “In general, would you say your mental health is…” with response options of “Excellent”, “Very good”, “Good”, “Fair” and “Poor”. This item has seen widespread use in the Canadian context [42,43].

Given prior documented associations between climate worry and depression [30], the Patient Health Questionnaire-8 (PHQ-8) [44] was also included, which measures depressive symptoms over the prior two weeks. The PHQ-8 has been used with youth samples [45], and was selected for this study as it omits an item on suicidal or self-injurious thoughts, while still being a valid measure of depressive symptoms [44]. For the PHQ-8, cut-offs were used to categorize participants as having no significant depressive symptoms (scores <5), mild depressive symptoms (scores ≥ 5 and <10), moderate depressive symptoms (scores ≥ 10 and <15), moderately severe depressive symptoms (scores ≥ 15 and < 20), or severe depressive symptoms (scores ≥ 20) [44]. The Cronbach’s-alpha for the PHQ-8 in this sample was 0.89.

Additionally, the 10-item version of the Kessler Psychological Distress Scale (K10) was used to measure psychological distress over the previous 30 days [46]. For the K10, cut-offs were used to categorize participants as having no/low psychological distress (scores ≤ 15), moderate psychological distress (scores >15 and ≤ 21), high psychological distress (scores >21 and ≤ 29), or very high psychological distress (scores ≥ 30) [47]. The Cronbach’s alpha for this scale was 0.93.

Finally, to explore impacts on functioning, the 36-item World Health Organization Disability Assessment Schedule-2.0 (WHODAS) [48] was used to assess functioning over the previous 30 days across six sub-domains: cognition, mobility, self-care, getting along with others, life activities, and participation. This scale has been previously validated among Canadian youth experiencing mental health challenges [49]. WHODAS scores for each of the sub-domains were calculated using the Simple Scoring Method [48], in which values for individual items are summed and divided by the number of items in the sub-domain. This scoring method is considered suited to clinical contexts [48] and process results in scores ranging from 1 to 5, with higher scores indicating greater difficulties with everyday functioning. The Cronbach’s alphas for the six sub-scales of the WHODAS were 0.86, 0.86, 0.75, 0.84, 0.94, and 0.93, respectively.

To assess climate worry and daily life impacts due to feelings about climate change, we used two separate items drawn from a prior 10-country survey of youth [11]; these items have also been used in research in Brazil [50] and with Canadian youth [13]. One item explored climate worry through the question: “I am worried that climate change threatens people and the planet”, with the response options including: “Not worried", "A little worried", "Moderately worried", "Very worried", "Extremely worried", and "Prefer not to say". Similar single-item measures of climate worry have been used in prior studies [29–31]. The other climate item assessed daily life impacts due to climate distress more generally: "My feelings about climate change negatively affect my daily life (at least one of the following is negatively affected: Eating, concentrating, work, school, sleeping, spending time in nature, playing, having fun, relationships)", with the response options, "yes", "no", and "prefer not to answer".

2.2. Analysis

Bivariate relationships between ordinal variables were examined using Goodman and Kruskal's gamma. Mann Whitney-U tests were conducted to compare median scores on the PHQ-8, K10, and WHODAS sub-scales for those who did and did not report any level of climate worry, those who did and did not report extreme climate worry, and those who did and did not report negative daily life impacts due to feelings about climate change. Given that prior research has found differences among youth based on age [18,51], age was dichotomized to compare older (19-24 years) and younger (16-18) youth. Chi-square tests were used to examine differences in the prevalence of reporting negative daily life impacts across sociodemographic characteristics; post-hoc comparisons using a Bonferroni correction were conducted as needed. All analyses were completed using SPSS 29.0 [52]. A p-value less than 0.05 was considered statistically significant and pairwise deletion was used to address missing data.

3. Results

3.1. Sociodemographic characteristics

A total of 393 youth participated in the survey. The mean age of participants was 20.4 years (standard deviation 2.58), and 48.1% identified as women. The majority of participants were living in market housing, although 9.9% were homeless or living in a shelter. Sociodemographic characteristics are presented in Table 1.

Table 1
Sociodemographic characteristics of the sample.

Age	n	%
16-18	101	25.7
19-24	292	74.3
Gender identity*	**n**	**%**
Woman	189	48.1
Gender diverse	108	27.5
Man	93	23.7
Missing	3	0.8
Ethnicity **	**n**	**%**
White/Caucasian (European)	177	45.0
First Nations/Métis/Inuit	73	18.6
East Asian (Chinese, Japanese, Korean)	64	16.3
Southeast Asian (Filipino, Indonesian, Thai)	35	8.9
Other (mixed ethnicity)	34	8.7
Hispanic/Latino (Chilean, Columbian, Mexican)	27	6.9
Middle Eastern/North African (Armenian, Egyptian, Iranian, Lebanese, Moroccan)	26	6.6
South Asian (Bangladeshi, Indian, Pakistani, Nepalese)	21	5.3
Black (Kenyan, Nigerian, Somali)	11	2.8
Caribbean (Haitian, Jamaican)	7	1.8
Prefer not to answer	8	2.0
Housing	**n**	**%**
Market rental (apartment with roommates, family home)	234	59.5
Homeless/shelter	39	9.9
Shared housing (where more than 5 people share a bathroom)	35	8. 9
Supportive housing (group home, support staff in the building)	27	6.9
Missing	58	14.8
Highest level of completed education	**n**	**%**
Less than high school	98	24.9
High school graduate	142	36.1
Some college	101	25.7
Undergraduate or graduate degree	52	13.2

* : Participants were asked "How would you describe your gender identity? (check all that apply)", and were provided the following response options: female, male, non-binary, Two-Spirit, Trans female, Trans male, not sure/questioning, I don't identify with any of these options, and Prefer not to answer. Participants who selected only male were categorized as man and participants who selected only female were categorized as woman. Participants who chose other options, or combinations of options, were considered gender diverse.

** : Participants were asked: "Which ethnicity do you identify with? (check all that apply)".

3.2. Mental health characteristics

Table 2 presents mental health characteristics of the sample. There was a high degree of mental health concern among participants, with over half screening positive for very high psychological distress. Over one in five were categorized as experiencing severe depressive symptoms.

3.3. Climate worry and daily life impacts due to emotions about climate change

As shown in Table 2, a total of 40.9% of participants reported being very or extremely worried about climate change, and 28.0% reported experiencing negative impacts on their daily lives due to their feelings about climate change. There was a strong positive relationship between level of climate worry and negative impacts to daily life due to feelings about climate change ($G=0.53$, $p<0.001$) (Figure 1). Notably, even among participants who reported no climate worry, 11.5% indicated that their feelings about climate change had a negative impact on their daily life. Among those who endorsed extreme climate worry, 34.3% reported that their feelings were *not* impacting their daily life.

3.4. Climate worry, daily life impacts due to emotions about climate change, and wellbeing

There was a weak negative relationship between climate worry and self-reported mental health ($G=-0.19$, $p<0.001$). However, there was no relationship between climate worry and depressive symptoms ($G=0.06$, $p=0.32$) or between climate worry and psychological distress ($G=0.13$, $p=0.07$). There was a moderate negative relationship between reporting negative daily impacts due to feelings about climate change and self-reported mental health ($G=-0.22$, $p=0.01$), as well as moderate positive relationships between reporting negative daily impacts due to feelings about climate change and both depressive symptoms ($G=0.21$, $p=0.02$) and psychological distress ($G=0.35$, $p<0.001$).

As shown in Table 3, the difference in median scores between those reporting any level of climate worry and those who reported not being worried was not statistically significant for the PHQ-8, K10, or any of the WHODAS subdomains. Compared to those who did not report extreme climate worry, individuals who did report extreme climate worry had higher median scores for the Cognition, Self-care, and Getting along subdomains of the WHODAS. Those who reported negative daily life impacts had higher median scores for the PHQ-8 and K10, as well as five of the sub-domains of the WHODAS, with the life activities sub-domain the only exception.

3.5. Sociodemographic variables and negative daily life impacts

The relative prevalence of reporting that feelings about climate

Table 2
Mental health, and climate distress characteristics of the sample.

Mental health characteristics		
Self-reported health	n	%
Excellent	29	7.4
Very good	77	19.6
Good	156	39.7
Fair	100	25.4
Poor	31	7.9
Self-reported mental health	n	%
Excellent	12	3.1
Very good	38	9.7
Good	83	21.1
Fair	146	37.2
Poor	114	29.0
Depressive symptoms (PHQ8)	n	%
No significant depressive symptoms	29	7.4
Mild depressive symptoms	68	17.3
Moderate significant depressive symptoms	74	18.8
Moderately severe depressive symptoms	83	21.1
Severe depressive symptoms	91	23.2
Missing/unable to calculate	48	12.2
Psychological distress (K10)	n	%
None to low psychological distress	19	4.8
Moderate psychological distress	38	9.7
High psychological distress	80	20.4
Very high psychological distress	210	53.4
Missing/unable to calculate	46	11.7
WHODAS sub-domain scores	Median	Interquartile range
Cognition (n=345)	2.33	1.17
Mobility (n=351)	1.60	1.20
Self-care (n=341)	1.75	1.25
Getting along with others (n=287)	2.20	1.60
Life activities (n=331)	2.50	1.38
Participation (n=293)	2.38	1.50
Climate distress characteristics		
I am worried that climate change threatens people and the planet	n	%
Not worried	29	7.4
A little worried	60	15.3
Moderately worried	133	33.8
Very worried	85	21.6
Extremely worried	76	19.3
Prefer not to say	9	2.3
Skipped question	1	0.3
My feelings about climate change negatively affect my daily life (at least one of the following is negatively affected: Eating, concentrating, work, school, sleeping, spending time in nature, playing, having fun, relationships)	n	%
No	247	62.8
Yes	110	28.0
Prefer not to answer	34	8.7
Skipped question	2	0.5

change negatively affect daily life by various sociodemographic variables is presented in Table 4. Those who were gender diverse had a higher prevalence of negative daily life impacts than those who identified as men or as women. Furthermore, individuals aged 19-24 years were more likely to report negative life impacts compared to those who were younger.

4. Discussion

Despite calls to better support youth experiencing mental health concerns related to climate change [22,53], there is limited available data from clinical contexts to provide evidence-based guidance. This paper, based on surveys of youth accessing services at one of four IYS centres in BC, presents some of the first quantitative data on climate worry and daily life impacts due to climate-related emotional responses among a clinical sample of youth.

Nearly all of the youth in this study reported experiencing climate worry to some degree, many at a high level. Further, over a quarter reported that their feelings about climate change were negatively impacting their daily lives. This appears lower than what has been reported in community-based population surveys of youth in Canada [13], although the reasons for this discrepancy are unclear. While there are limited clinical data to make comparisons, research conducted at a youth outpatient mental health practice in the United States found a high level of mental health concern related to climate change, with approximately two-thirds of participants reporting anxiety due to hearing about climate change, with symptoms including worry, anxiety about the future, and difficulty sleeping [54].

In the present study, there was a strong positive relationship between level of climate worry and reporting that one’s feelings about climate change negatively impact daily life. However, the associations between climate worry and measures of mental health and functioning were

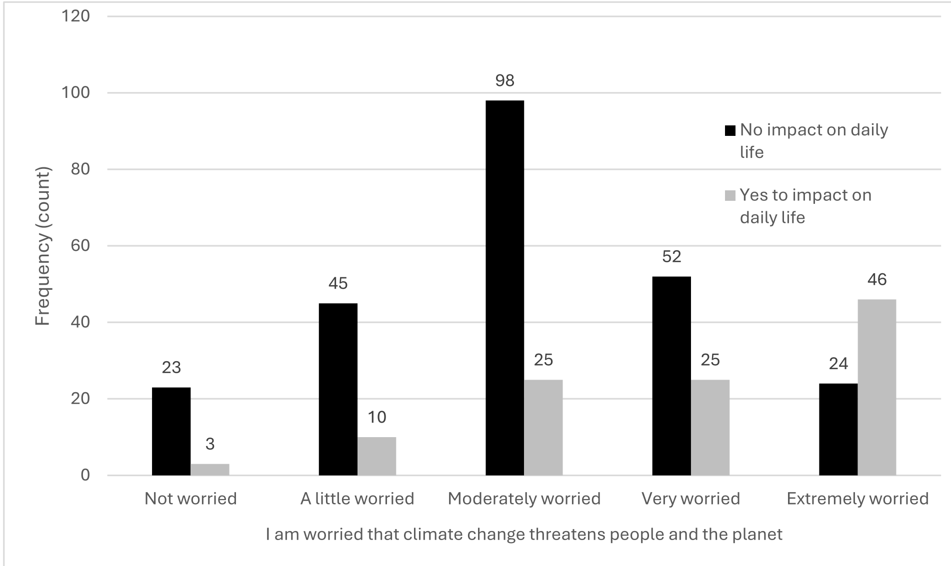

Figure 1. Relationship between climate worry and reporting that feelings about climate change negatively impact daily life.

Table 3
Comparison of median scores on the PHQ-8, K10, and WHODAS sub-scales by climate distress responses*.

Scale or sub-scale	Comparison 1			Comparison 2			Comparison 3		
	Any climate worry **	No climate worry		Extreme climate worry	Not extreme climate worry ***		Daily life impacts	No daily life impacts	
	Median	Median	p-value	Median	Median	p-value	Median	Median	p-value
PHQ-8	15.00	14.00	0.26	16.00	14.00	0.06	**16.00**	**14.00**	**0.007**
K10	32.00	33.00	0.85	34.00	32.00	0.08	**35.00**	**30.00**	**<0.001**
WHODAS: cognition	2.33	2.33	0.99	**2.83**	**2.33**	**0.01**	**2.83**	**2.17**	**<0.001**
WHODAS: mobility	1.60	1.70	0.73	1.90	1.60	0.36	**2.00**	**1.60**	**0.01**
WHODAS: self-care	1.75	1.75	0.80	**2.13**	**1.75**	**0.01**	**2.00**	**1.75**	**0.004**
WHODAS: getting along with others	2.20	2.00	0.61	**2.40**	**2.00**	**0.03**	**2.40**	**2.00**	**0.008**
WHODAS: life activities	2.38	2.56	0.72	2.81	2.38	0.19	2.75	2.38	0. 31
WHODAS: participation	2.38	2.88	0.17	2.63	2.38	0.08	**2.75**	**2.25**	**<0.001**

* : Mann Whitney-U tests used to compare median scores, p-values <0.05 are bolded.
** : Those who reported being “A little worried”, “Moderately worried”, “Very worried”, or “Extremely worried”.
*** : Those who reported being "Not worried”, “A little worried”, “Moderately worried”, or “Very worried”.
K10: Kessler Psychological Distress Scale, PHQ-8: Patient Health Questionnaire-8, WHODAS: World Health Organization Disability Assessment Schedule-2.0

inconsistent, with only a weak relationship with self-reported mental health, and unsignificant associations with depressive symptoms and psychological distress. These findings support the notion that climate distress may not be inherently pathological [11,20,21], but contrast with other research documenting significant relationships between climate worry and depression [30,31].

What seems important, from a clinical perspective, is reporting impacts on daily life as a result of negative climate emotions. Youth who reported that their emotions about climate change negatively impacted their daily life had higher median scores on the PHQ-8, K10, and on five of the six sub-domains of the WHODAS. Given the associations with worsened wellbeing across multiple domains, reporting such impacts likely indicates the need for further assessment. This is in line with the view that impacts to daily functioning warrant clinical attention [35]. However, while there have been suggestions that clinicians screen for the impact of climate change on youth mental health [33], none of the existing scales have been created for this purpose [18].

Among youth who did not report any worry about climate change, a small number still indicated experiencing negative impacts to their daily lives because of their feelings about climate change. This suggests that worry is not the only emotional response to climate change associated with mental health concerns. Youth whose climate distress manifests in other feelings, such as grief, risk being “missed” if questions focus on worry alone. Open-ended questions may thus be of use during clinical assessment [18]. Conversely, even among those who reported being extremely worried about climate change, one in three did not endorse negative daily life impacts. This suggests that youth can potentially cope with even high levels of climate worry, supporting the notion that strong emotions in response to climate change do not necessarily manifest in a clinically significant manner. While there have been numerous studies into how youth cope with the emotions brought on by climate change [55–57], to our knowledge, none are focused on clinical contexts.

Our findings further show that the prevalence of reporting daily life impacts due to climate change varies by sociodemographic characteristics. Specifically, youth who are gender diverse are more likely to report negative daily life impacts due to climate change. This finding is consistent with prior population-level research in Canada [12,14,28] and elsewhere [58]. While the underlying reasons are not clear, it is important to note that gender diverse youth experience a number of mental health inequities [47,59]. Findings indicating that older youth are more likely to experience negative impacts to daily functioning as a result of climate change-related emotions also align with

Table 4
Prevalence of reporting that feelings about climate change negatively affect daily life by sociodemographic characteristics.

	My feelings about climate change negatively affect my daily life		
	n	%	p-value *
Gender identity **			
Woman (n=177)	44	24.9[a]	**<0.001**
Gender diverse (n=98)	45	45.9[b]	
Man (n=80)	20	25.0[a]	
Education level			
Not completed high school (n=86)	20	23.3	0.08
Complete high school or more (n=271)	90	33.2	
Housing situation			
Market rental (apartment with roommates, family home) (n=217)	63	29.0	0.51
Homeless/shelter (n=35)	13	37.1	
Shared housing (where more than 5 people share a bathroom) (n=32)	12	37.5	
Supportive housing (group home, support staff in the building) (n=22)	5	22.7	
Age			
16-18 (n=92)	20	21.7	**0.03**
19-24 (n=265)	90	34.0	

* : Determined via Chi-square. Superscript letters for the gender variable indicate which categories differ at the $p<0.05$ level, based on post-hoc analysis, using a Bonferroni correction.

** : Participants were asked "How would you describe your gender identity? (check all that apply)", and were provided the following options: female, male, non-binary, Two-Spirit, Trans female, Trans male, not sure/questioning, I don't identify with any of these options, and Prefer not to answer. Participants who selected only male were categorized as man and participants who selected only female were categorized as woman. Participants who chose other options, or combinations of options, were considered gender diverse.

population-level research into age and climate distress [18].

This study has important limitations. To start, the research is based on a convenience sample from a limited number of IYS centres in BC, Canada. As such, caution must be taken when generalizing to youth in other contexts or accessing different kinds of clinical services. This is particularly the case, given that none of the four centres in the study were located in rural areas; prior research has indicated that there may be differences in youth climate distress across urban and rural areas [12].

Another limitation is that of missing data. The question on housing status was only added partway through data collection, resulting in a relatively high amount of missingness and reducing statistical power. Furthermore, it was not possible to calculate PHQ-8, K10, or WHODAS scores for all youth. The fact that entire scales were often skipped, along with all subsequent questions, suggests that some youth may have been leaving surveys incomplete after being called to appointments. One exception is an item on the WHODAS related to sexual activity. This item had a higher degree of missingness than others, suggesting that youth may have been choosing not to respond to this question. This issue of missingness may be related to another limitation of the study, which was the relatively long length of the survey.

Finally, there is the issue of causation. Indeed, the difficulty of disentangling causal pathways between various forms of climate distress and more general mental health concerns has been identified as a widespread issue [5]. While climate change may be driving mental health concerns in some youth, it is also possible that existing mental health challenges predispose some youth to climate distress, or that causal pathways are bidirectional [26]. However, regardless of the causal direction, the clinical implications are clear: if a youth reports that their emotions about climate change are negatively impacting their life, further follow-up is needed to guide related supports.

These findings suggest several areas for future research. For example, there is a need for research exploring how youth who report extreme climate worry, but no impact to their daily lives, differ from those who identify extreme worry alongside daily life impacts. Such differences could reveal protective factors and potential interventions. Further research is also needed to better understand the underlying pathways connecting climate distress and socio-demographic characteristics, such as gender. More broadly, given the limited data, it would be helpful to examine this issue in other, varied clinical contexts.

5. Conclusion

This study is among the first to examine climate change and mental health among youth accessing clinical services. Many of the youth were very or extremely worried about climate change, and a notable subset identified that their feelings about climate change were negatively impacting their daily lives, suggesting a need for clinical support. These results underscore the urgent need for service providers to be attuned to the impacts of climate change on youth mental health, and for further research to refine assessment methods for these impacts, determine protective factors, and understand what forms of support youth find most helpful.

Data availability statement

The datasets generated and/or analysed during the current study are not publicly available due to privacy but are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Zachary Daly: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Emily K. Jenkins:** Writing – original draft, Supervision, Methodology, Conceptualization. **Sarah Adair:** Writing – review & editing, Project administration, Investigation, Data curation. **A. Fuchsia Howard:** Writing – review & editing. **Skye Barbic:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

Zachary Daly serves on the board of Canadian Association of Nurses for the Environment (CANE) and has accepted speaking fees from the Canadian Association of Physicians for the Environment (CAPE). Neither organization had any role in the collection of data, the writing of the manuscript, or the decision to submit for publication. Other authors declare that they have no competing or potential conflicts of interest.

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